

Statistical Methods in AI

Assignment 3: Build a Real ML App

13 Themes · 103 Variants · Pick One

IIIT Hyderabad · Academic Year 2025–26

Read this catalogue, discuss with your team, and submit your top 3 preferences via the assignment form.

Time budget: 4–6 hours per person (total).

Compute: Google Colab free tier (T4 GPU) or any laptop. No paid GPUs needed.

Deliverables: (i) a working web app (Streamlit / Gradio / HF Space) (ii) a 4–6 page report (iii) GitHub repo.

Tier 1: 1–3 students. Smaller dataset, simpler scope, lighter report.

Tier 2: 5–7 students. Larger dataset, deeper evaluation, ICVGIP-style report.

Use of LLMs (ChatGPT, Claude, Gemini) for data labelling, code scaffolding, and report drafting is encouraged.

You must disclose what you used and explain every line of your own code in the viva.

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Before You Pick

Why this assignment exists

Every topic here:

- Solves a problem a non-CS person would understand.
- Has a clear, accessible dataset (link given for every variant).
- Ends in a demo you can share on LinkedIn.
- Fits in the time budget without burning your week.

What you will hand in

1. **GitHub repository.** Public, with `README.md`, `requirements.txt`, training notebook, app code.
2. **Live demo.** Deployed on HuggingFace Spaces (free) or Streamlit Community Cloud (free). If hosting fails, a recorded ≤ 2 -minute screen demo is acceptable.
3. **Technical report (PDF).** Sections: Introduction, Data, Method, Results, Limitations, References. Include an Ablation section where necessary.
4. **One-slide pitch.** Single PNG/PDF you could post on LinkedIn.

The two tiers

	Tier 1 (easier)	Tier 2 (more involved)
Team size	1–3 students	5–7 students
Per-person time	4–6 hours	4–6 hours
Dataset scale	200–2,000 examples	2,000–20,000 examples
Models	Pre-trained / zero-shot / 1-epoch fine-tune	Light fine-tune + ablation + comparison
Report	4–6 pages, lab-report style	6–8 pages, ICVGIP-style
App polish	Working Streamlit demo	Polished + deployed on HF Spaces
Maximum marks	100	100

Tier 2 is not “worth more marks” — the ceiling is the same. Tier 2 simply asks more from a larger team. It may be more fun and engaging and would generally result in deeper insights and learnings.

Using LLMs (encouraged!)

You may freely use ChatGPT / Claude / Gemini / Copilot for: data labelling (paste 50 reviews, ask for sentiment labels), synthetic data generation, code scaffolding (Streamlit UI, dataset loaders), debugging, and report drafting. **Honesty rules:** (a) cite the LLMs you used in Acknowledgements, (b) all evaluation and analysis must be your own.

How to choose and submit

1. Read the 13 themes. Skim variants in the ones that interest you.
2. Form your team (pick a tier first — it sets the size).

3. Submit via the Google Sheet (https://docs.google.com/spreadsheets/d/1E6P61o_E4S-YG61YbZmCb-s1nJa5bUCRkLq4w/edit?usp=sharing). Only select from the available ones which have not already been taken (from the google sheet) - first come first serve basis.
4. **Want to propose your own project?** The form has an optional section. Faculty/TA approval required.

T10 — Government-Services RAG Chatbot

The problem. Indian government services (passport, voter ID, RTI, taxes, etc.) have lots of public PDF documentation. Build a chatbot that answers “how do I apply for X” by retrieving from official PDFs.

Why this is the easiest theme. No model training at all. You build a RAG (Retrieval-Augmented Generation) pipeline using free embeddings + a free LLM API.

What you will build. Streamlit chat (`st.chat_input`). Each answer shows source PDF citations. Optional: “Talk in Hindi” toggle.

Skills. Document parsing, embeddings, vector search, prompt engineering, chat UI.

Common base.

- Embeddings: `sentence-transformers/all-MiniLM-L6-v2` (free, CPU).
- Vector store: `chromadb` or `faiss-cpu`.
- LLM: free Google AI Studio (Gemini 1.5 Flash) or Groq (Llama-3 free).

Variants

ID	Variant	Tier	Source PDFs (the “dataset” is the PDF set you download once)
T10.1	Passport assistant	Tier 1	https://www.passportindia.gov.in (download 5–10 “Instruction Booklet” / FAQ PDFs).
T10.2	Voter ID / EPIC assistant	Tier 1	Election Commission of India: https://www.eci.gov.in (Form-6/7/8 instruction PDFs).
T10.3	RTI filing helper	Tier 1	https://rti.gov.in (RTI Act PDF + state-rules PDFs).
T10.4	Income tax FAQ chatbot	Tier 1	https://www.incometax.gov.in (ITR forms + instruction booklets).
T10.5	Driving licence & RC	Tier 1	https://parivahan.gov.in (Sarathi/Vahan PDFs).
T10.6	Ayushman Bharat eligibility	Tier 1	National Health Authority: https://nha.gov.in (PM-JAY guidelines PDFs).
T10.7	Multi-service unified + Hindi support	Tier 2	5+ of the above sources combined; language toggle, conversation memory.
T10.8	IIIT-H rules / hostel / academic bot	Tier 2	Public IIIT-H academic regulations + hostel rules PDFs from https://www.iiit.ac.in .